

Border Subgrid Count II

Time limit: 1.00 s

Memory limit: 512 MB

You are given a grid of letters. Your task is to calculate, for each letter, the number of *rectangular* subgrids whose border consists of that letter.

Input

The first line has two integers n and k : the size of the grid and the number of letters. The letters are the first k uppercase letters.

After this, there are n lines that describe the grid. Each line has n letters.

Output

Print k lines: for each letter, the number of subgrids.

Constraints

- $1 \leq n \leq 500$
- $1 \leq k \leq 26$

Example

Input	Output
5 3 ABBBC ABABC ABBBC ABBBC CCCCC	11 38 29